

Business Insights and Recommendations Report**Heart Failure Prediction — Advanced Data Analysis & Statistical Validation****Executive Summary**

Cardiovascular disease remains one of the leading causes of death globally, and early identification of at-risk patients using accessible clinical measurements can support earlier intervention and better resource allocation. This report summarizes the advanced analysis, statistical validation, and feature engineering conducted on a dataset of 918 patient records as part of Week 3 of the AnalystLab Africa Data Science Internship.

The analysis confirms multiple clinically meaningful predictors of heart disease. Four statistical tests were conducted, all returning statistically significant results ($p < 0.05$). A newly engineered composite Risk Score outperformed every original feature in its relationship to the target variable, and several counterintuitive findings — most notably around chest pain presentation — carry direct implications for how heart disease risk is screened and communicated.

The refined dataset produced through this analysis is ready to serve as the foundation for predictive model development in Week 4.

Key Analytical Findings**Statistically Significant Relationships**

Four hypothesis tests were conducted to validate relationships suggested during exploratory analysis. All four returned statistically significant results:

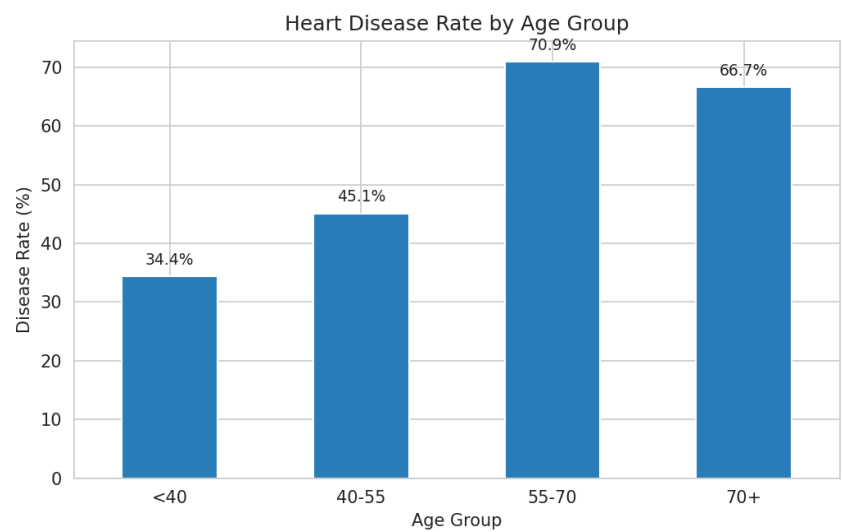
Test		Variables	Result	Interpretation
Welch's t-test		MaxHR × HeartDisease	$t = 13.23, p < 0.000001$	No-disease patients average 148.2 bpm vs. 127.7 bpm for disease-positive patients
Chi-Square		ChestPainType × HeartDisease	$\chi^2 = 268.07, p < 0.000001$	Strong association; asymptomatic patients show far higher disease prevalence
Mann-Whitney U		Cholesterol × ExerciseAngina	$U = 112,811, p = 0.0039$	Statistically significant but practically weak — identical medians (237) in both groups
One-Way ANOVA		MaxHR × ST_Slope	$F = 79.03, p < 0.000001$	"Up" slope patients average 148.0 bpm vs. ~128-130 bpm for "Flat"/"Down"

Chest Pain Presentation: A Counterintuitive Finding

The single most notable finding of this analysis concerns the type of chest pain. Patients presenting with no chest pain symptoms (classified as "Asymptomatic") showed a disease rate of 79%, compared to only 14% among patients with Atypical Angina. This is counterintuitive: the absence of a classic symptom is associated with substantially higher, not lower, disease prevalence.

Age-Related Risk Gradient

Disease prevalence rises consistently with age, from 34.4% in patients under 40 to 70.9% in the 55-70 age band, before a slight decline in the 70+ group (likely reflecting a smaller sample in that band rather than a genuine risk reduction).



Feature Engineering & Evaluation Insights

Three features were engineered based on findings from the statistical analysis: an Age Group band, a Cholesterol-to-Age ratio, and a composite Risk Score combining three individually significant clinical indicators (exercise-induced angina, fasting blood sugar, and above-median ST depression).

The composite Risk Score proved to be the strongest predictor identified in this analysis — stronger than any original clinical feature considered individually.

Feature	Correlation with Target	Status
Risk Score (engineered)	0.577	Retained: strongest predictor
Oldpeak	0.404	Retained
MaxHR	-0.400	Retained
Age	0.282	Retained
Fasting Blood Sugar	0.267	Retained
Resting BP	0.118	Retained
Cholesterol-Age Ratio (engineered)	-0.114	Removed: high multicollinearity, weak signal

Feature	Correlation with Target	Status
Cholesterol	0.076	Retained despite weak standalone signal

The Cholesterol-to-Age Ratio feature was evaluated and ultimately removed from the final modelling dataset. While conceptually reasonable, it showed a Variance Inflation Factor of approximately 20, indicating substantial multicollinearity with its own source variables (Age and Cholesterol) — combined with a weak standalone correlation to the target. Retaining it would have added redundancy without proportional analytical value.

Business Implications

- Screening protocols that rely heavily on symptomatic presentation (chest pain) risk under-detecting a substantial share of at-risk patients who present asymptotically.
- A composite, multi-indicator risk-scoring approach appears meaningfully more informative than relying on any single clinical measurement, including cholesterol — a variable often assumed to be a primary risk indicator but which showed the weakest standalone relationship to the target in this dataset.
- Maximum heart rate achieved during exercise testing and ST segment slope both show strong, statistically robust separation between disease-positive and disease-negative patients, supporting their continued clinical relevance.

Recommendations

- Prioritize the composite Risk Score, ST_Slope, MaxHR, Oldpeak, and ChestPainType as core features in Week 4 predictive modelling, given their demonstrated statistical and correlational strength.
- Exclude the Cholesterol-to-Age Ratio feature from linear modelling approaches due to confirmed multicollinearity; it may still be considered for tree-based models less sensitive to this issue, if interpretability value is desired.
- Treat cholesterol-related findings with caution in stakeholder communication — while clinically important in general medical literature, their predictive value within this specific dataset was comparatively weak.
- Communicate the asymptomatic-risk finding to relevant stakeholders, as it has direct implications for how screening criteria are framed.

Dataset Limitations

- **Sex imbalance:** approximately 79% of records are male, which limits the generalizability of any model trained on this data to female patients.
- **Sample size:** 918 records is a moderate sample for clinical generalization; findings should be validated against larger, more diverse cohorts before any real-world screening application.
- **Statistical vs. practical significance:** the Cholesterol × ExerciseAngina test returned a statistically significant result ($p = 0.0039$) despite near-identical medians between groups, illustrating that statistical significance does not always imply meaningful practical effect size — a distinction that should guide, not override, clinical interpretation.
- **Cross-sectional data:** this dataset captures a single point in time per patient and cannot establish causality between clinical measurements and heart disease onset.

Next Steps

The refined dataset produced through this analysis — with the Cholesterol-to-Age Ratio removed and three validated engineered features retained — is ready to serve as the

foundation for Week 4: Machine Learning Model Development, Evaluation & Business Recommendations. The feature relationships and statistical evidence documented in this report provide a grounded starting point for feature selection and model interpretation in that next phase.